Scalability & Future Plan

Date	1 July 2026
Team ID	XXXXXX
Project Name	**OptiCrop – Crop Recommendation System**
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application relies on manually entered soil and environmental parameters.	Users must have soil test data before using the system.	High
2	The current system predicts only the most suitable crop and does not provide fertilizer or irrigation recommendations.	Limits decision-making support for farmers.	Medium
3	The application does not integrate real-time weather or soil sensor data.	Predictions depend entirely on user-provided values.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Suitable for a limited number of users.	Deploy on scalable cloud infrastructure with load balancing and auto-scaling.
2	Data Storage	Uses a static CSV dataset and trained model without a database.	Integrate a relational database (MySQL/PostgreSQL) to store user data, prediction history, and analytics.
3	Performance	Processes one prediction request at a time.	Optimize model inference, implement caching, and support concurrent user requests.
4	Security	Basic deployment without user authentication.	Add secure login, HTTPS enforcement, role-based access control, and encrypted data storage.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact

Phase 1	Improve the user interface with modern responsive design and enhanced user experience.	1–2 Months	Easier navigation and improved usability.
Phase 2	Integrate real-time weather API and fertilizer recommendation system.	2–3 Months	More accurate crop recommendations and better farming decisions.
Phase 3	Add multilingual support, user authentication, and prediction history.	3–6 Months	Increased accessibility and personalized user experience.
Phase 4	Integrate IoT soil sensors, mobile application support, and cloud database.	6–12 Months	Fully automated smart agriculture solution with real-time monitoring and recommendations.